



HK IMO Training 2018-2019

Problem Set 9

Date due: 1 December 2018



Problem 37. Let $n \geq 3$ be an integer. What is the largest possible number of interior angles greater than 180° in an n -gon in the plane, given that the n -gon does not intersect itself and all its sides have the same length?

Problem 38. In a country there are 2018 airports, no three of them lying on a line. Two airports are connected by a direct flight if and only if the line passing through them divides the country in two parts, each with 1008 airports in it. Show that there are no two airports such that one can travel from the first to the second, visiting each of the 2018 airports exactly once.

Problem 39. Let $ABCD$ be a convex quadrilateral such that $BC = AD$. Let M and N be the midpoints of AB and CD , respectively. The lines AD and BC meet the line MN at P and Q , respectively. Prove that $CQ = DP$.

Problem 40. Find all functions $f: \mathbb{R} \rightarrow \mathbb{R}$ for which

$$f(x+y) = f(x-y) + f(f(1-xy))$$

holds for all real numbers x and y .